Resolving the Remaining Cases of Carlitz-Extensiveness over Finite Fields using SageMath

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Main goal of this talk

The question

For which finite field extensions $\mathbb{F}_{q^n}/\mathbb{F}_q$ does there exist, for *every* $z \in \mathbb{F}_{q^n}$, an element $x \in \mathbb{F}_{q^n}$ that is both primitive *and* a γ_z-generator?

- Open since Hsu & Nan (2011); narrowed by Hachenberger (2016).
- Six pairs (q, n) remained unresolved.

Lenstra–Schoof	Hsu & Nan	Hachenberger	Gruber	this work
1987	2011	2016	2016	2026

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What we show

Using SageMath, we settle the six remaining cases. The complete classification:

$$\mathbb{F}_{q^n}/\mathbb{F}_q \text{ is Carlitz-extensive} \iff (q, n) \neq (2, 2).$$

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The Primitive Normal Basis Theorem

Primitive and normal elements

Let $\mathbb{F}_{q^n}/\mathbb{F}_q$ be a finite field extension of degree n.

Primitive element ($\rightarrow$ multiplicative structure)

$\alpha \in \mathbb{F}_{q^n}$ is **primitive** if $\mathbb{F}_{q^n}^\times = \langle \alpha \rangle$. Equivalently, $\mathrm{ord}(\alpha) = q^n - 1$.

Normal element ($\rightarrow$ additive structure)

$\alpha \in \mathbb{F}_{q^n}$ is **normal** over $\mathbb{F}_q$ if the iterates of the Frobenius automorphism form an $\mathbb{F}_q$-basis of $\mathbb{F}_{q^n}$:

$$\{\alpha,\ \alpha^q,\ \alpha^{q^2}, \ldots, \alpha^{q^{n-1}}\} \quad \text{is an } \mathbb{F}_q\text{-basis of } \mathbb{F}_{q^n}.$$

The Primitive Normal Basis Theorem (PNBT)

Theorem (Lenstra & Schoof, 1987)

For every finite field extension $\mathbb{F}_{q^n}/\mathbb{F}_q$ there exists an element $\alpha \in \mathbb{F}_{q^n}$ which is simultaneously primitive and normal over $\mathbb{F}_q$.

Such an α generates $\mathbb{F}_{q^n}^{\times}$ multiplicatively and $\mathbb{F}_{q^n}$ additively (via its Frobenius basis).

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The Hsu–Nan Generalisation (2011)

The endomorphism γ_z

Hsu & Nan (2011) generalise the Primitive Normal Basis Theorem by replacing the Frobenius automorphism σ with a wider family of $\mathbb{F}_q$-linear maps.

Definition

For $z \in \mathbb{F}_{q^n}$, define

$$\gamma_z\colon \mathbb{F}_{q^n} \longrightarrow \mathbb{F}_{q^n}, \qquad \gamma_z(x) \;=\; z \cdot x \;+\; x^q.$$

- γ_z is an $\mathbb{F}_q$-linear endomorphism of $\mathbb{F}_{q^n}$.
- Special case $z = 0$: $\gamma_0(x) = x^q = \sigma(x)$ is the Frobenius automorphism.

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Lemma (Minimal polynomial, Hsu–Nan)

If $z \in \mathbb{F}_{q^n}$ has minimal polynomial $f_z(x) \in \mathbb{F}_q[x]$ of degree $d \mid n$, then

$$\mathrm{m}_{\gamma_z}(x) \;=\; f_z(x)^{\,n/d} \;-\; 1.$$

γ_z-generators and Carlitz-extensiveness

Definition (γ_z-generator)

An element $\alpha \in \mathbb{F}_{q^n}$ is a γ_z-**generator** if

$$\{\alpha,\ \gamma_z(\alpha),\ \gamma_z^2(\alpha), \ldots, \gamma_z^{n-1}(\alpha)\} \quad \text{is an } \mathbb{F}_q\text{-basis of } \mathbb{F}_{q^n}.$$

- A γ_z-generator always exists but does not need to be primitive.

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- A γ_z-generator always exists but does not need to be primitive.

Definition (Carlitz-extensiveness)

$\mathbb{F}_{q^n}/\mathbb{F}_q$ is **Carlitz-extensive** if for every $z \in \mathbb{F}_{q^n}$ there exists a primitive γ_z-generator.

Counterexample: $\mathbb{F}_4/\mathbb{F}_2$ is not Carlitz-extensive

Setup. $\mathbb{F}_{q^n} = \mathbb{F}_{2^2} = \mathbb{F}_4 = \mathbb{F}_2[x]/(x^2+x+1)$, $\mathbb{F}_q = \mathbb{F}_2$. α root of x^2+x+1, so $\alpha^2 = \alpha+1$.

- $|\mathbb{F}_4^\times| = 3$ is prime $\implies$ primitive elements are $\{\alpha,\ \alpha+1\}$
- x is a γ_z-generator $\iff$ $\{x,\ \gamma_z(x)\}$ is an $\mathbb{F}_2$-basis, where $\gamma_z(x) = z\cdot x + x^2$

z	$\{\alpha,\ \gamma_z(\alpha)\}$	$\{\alpha+1,\ \gamma_z(\alpha+1)\}$
0	$\{\alpha,\ \alpha+1\}$ ✓	$\{\alpha+1,\ \alpha\}$ ✓
1	$\{\alpha,\ 1\}$ ✓	$\{\alpha+1,\ 1\}$ ✓
α	$\{\alpha,\ 0\}$ ✗	$\{\alpha+1,\ \alpha+1\}$ ✗
$\alpha+1$	$\{\alpha,\ \alpha\}$ ✗	$\{\alpha+1,\ 0\}$ ✗

Conclusion: For $z \in \{\alpha,\ \alpha+1\}$, only 1 but no primitive element is a γ_z-generator

Hsu & Nan: open cases

Hsu & Nan worked with character-sum estimates and solved all but 63 pairs (q, n) pairs. These 63 pairs were too small for asymptotic methods to apply.

Known exception

$(q, n) = (2, 2)$, i.e. $\mathbb{F}_4/\mathbb{F}_2$, is *not* Carlitz-extensive. Is it the only one?

Hsu & Nan, J. Number Theory 131 (2011), 146–157.

(Subsequent work reduced the 63 open cases to just six — as we will see.)

The Hachenberger Generalisation (2016)

Hachenberger: cyclic vector spaces

Hachenberger (2016) embeds the problem in a wider framework: replace γ_z by *any* $\mathbb{F}_q$-linear endomorphism τ of $\mathbb{F}_{q^n}$ for which a τ-generator exists (analogue to γ_z-generator). Such a τ is then called cyclic.

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Definition (Extensiveness)

$\mathbb{F}_{q^n}/\mathbb{F}_q$ is **extensive** if for every cyclic $\mathbb{F}_q$-endomorphism τ of $\mathbb{F}_{q^n}$ there exists a primitive τ-generator $\alpha \in \mathbb{F}_{q^n}$, i.e. α generates both $\mathbb{F}_{q^n}^\times$ and $\mathbb{F}_{q^n}$ as an $\mathbb{F}_q[\tau]$-module.

- Hachenberger solved the question of extensivity for all but 19 pairs (q, n).
- Specialisation: $\tau = \gamma_z$, $z \in \mathbb{F}_{q^n}$ $\Rightarrow$ primitive γ_z-generator $\to$ Hsu–Nan

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Progress on the 63 open cases:

- Hachenberger resolved 41 of them.
- Master's thesis of Thomas Gruber reduced the list by another 16 pairs.

Hachenberger, Australasian J. Combin. 64 (2016), 289–326.

The remaining 6 cases

After Hachenberger and Gruber, only six pairs (q, n) remained unresolved:

q	n
2	18, 20, 24
4	10, 12
8	8

- Too large for naive brute force.
- Too small for asymptotic / structural arguments.

Goal of this work: settle each case by an explicit computation in SageMath.

Our Approach: Reductions & Algorithm

Strategy at a glance

We reduce the search using:

1. Cyclotomic cosets: one z per orbit (corresponding to coset) suffices (outer loop), simplifies the calculation of the minimal polynomial f_z of z.

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Strategy at a glance

We reduce the search using:

1. Cyclotomic cosets: one z per orbit (corresponding to coset) suffices (outer loop), simplifies the calculation of the minimal polynomial f_z of z.
2. Projective space: one x per projective line suffices (inner loop), restrict the search for a primitive γ_z-generator to one representative per projective primitive line.
3. Minimal-polynomial factorisation: fast algebraic test for the γ_z-generator property, more efficient than e.g. the basis check.

Each tool decreases the number of computations needed, making the algorithm feasible.

Cyclotomic cosets and Frobenius compatibility

Cyclotomic coset (modulo $q^n - 1$)

$C_k = \{k, kq, kq^2, \dots\} \pmod{q^n - 1}, \ |C_k| \mid n.$

For a primitive $g \in \mathbb{F}_{q^n}$, exponentiation turns C_k into the Frobenius orbit of g^k, and yields the minimal polynomial $f_{g^k} = \prod_{j \in C_k}(x - g^j)$ easily.

Lemma (Conjugation lemma)

For every $z \in \mathbb{F}_{q^n}$ and $i \in \mathbb{Z}$, $\gamma_{\sigma^i(z)} = \sigma^i \circ \gamma_z \circ \sigma^{n-i}$.

- If α is a primitive γ_z-generator, then $\sigma^i(\alpha)$ is a primitive $\gamma_{\sigma^i(z)}$-generator.
- Reduction: test only one z per Frobenius orbit.

Projective space and $\mathbb{F}_q^\times$-invariance

Set $N = q^n - 1$ and $Q = N/(q-1)$. The projective $(n-1)$-space:

$$\mathbb{P}^{n-1}_{\mathbb{F}_q} = \mathbb{F}_{q^n}^\times / \mathbb{F}_q^\times \quad \text{(cyclic of order } Q\text{)}.$$

A projective point $[g^k]$ is primitive if it generates this quotient, i.e. $\gcd(k, Q) = 1$.

Lemma ($\mathbb{F}_q^\times$-invariance)

Let $z, x \in \mathbb{F}_{q^n}$ and $\lambda \in \mathbb{F}_q^\times$. Then

$$x \text{ is a } \gamma_z\text{-generator} \iff \lambda x \text{ is a } \gamma_z\text{-generator}.$$

Reduction: the γ_z-generator property depends only on the projective point $[x]$. Once a primitive projective point that is a γ_z-generator is found, some scalar multiple λx is automatically a primitive γ_z-generator.

The algorithm in one slide

Input. A pair (q, n). Output. `True` / `False` for Carlitz-extensiveness.

Preparation.

1. Fix a primitive element $g \in \mathbb{F}_{q^n}^\times$; set $N = q^n - 1$, $Q = N/(q-1)$.
2. Precompute representatives of the primitive projective points:
$$\mathcal{R} = \{\, g^k : 1 \le k < Q,\ \gcd(k, Q) = 1 \,\}.$$
3. Handle $z = 0$ separately (Frobenius case): search $\mathcal{R}$ for a primitive γ_0-generator.

Main loop. For each q-cyclotomic coset of $\mathbb{Z}/N\mathbb{Z}$:

1. Pick representative $z = g^e$ of Frobenius orbit.
2. Compute the minimal polynomial f_z of z via its cyclotomic coset, then factor $\mathrm{m}_{\gamma_z} = f_z^{n/\deg f_z} - 1 = \prod_i p_i^{e_i}$.
3. Build the matrix A of γ_z, then $H_i = (\mathrm{m}_{\gamma_z}/p_i)(A)$ for each i.
4. Scan $\mathcal{R}$: if $\mathrm{Vec}(g^k) \cdot H_i \neq 0$ for *all* i, then $\alpha = \lambda g^k$ is a primitive γ_z-generator.
5. If $\mathcal{R}$ is exhausted with no hit: return `False`.

If every coset succeeds: return `True`.

Results

Main result

Theorem (Hellwig & Schaller, 2026)

Each of the six remaining pairs

$$(q, n) \in \{(2, 18), (2, 20), (2, 24), (4, 10), (4, 12), (8, 8)\}$$

is Carlitz-extensive.

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Combined classification

A finite-field extension $\mathbb{F}_{q^n}/\mathbb{F}_q$ is Carlitz-extensive if and only if $(q, n) \neq (2, 2)$.

Equivalently, $\mathbb{F}_4/\mathbb{F}_2$ is the unique non-Carlitz-extensive extension among all finite extensions of finite fields.

Verification and code

Two independent searches

- Original algorithm (projective-point first): $\approx$ 18h on one core for all six cases together.
- Alternative algorithm (generator-first): find one γ_z-generator u, then search candidates $a(\gamma_z)(u)$ with $a \in \mathbb{F}_q[x]$ monic, $\deg a < n$, $\gcd(a, \mathrm{m}_{\gamma_z}) = 1$. $\approx$ 21h on one core.

Both produce identical results on all six cases.

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Repository

Code, pseudocode, benchmarks, and additional material:

`github.com/florian-hellwig/Carlitz-extensiveness`

Outlook: Generalisations and Conjectures

$\mathbb{F}_q[x]$-Drinfeld extensiveness: an intermediate framework

Let $\tau\colon x \mapsto x^q$ be the Frobenius on $\mathbb{F}_{q^n}$.

Definition ($\mathbb{F}_q[x]$-Drinfeld endomorphism of rank r)

For $t, c_1, \ldots, c_r \in \mathbb{F}_{q^n}$ with $c_r \neq 0$,

$$\varphi \;=\; t \cdot \tau^0 + c_1\tau + \cdots + c_r\tau^r \;\in\; \mathrm{End}_{\mathbb{F}_q}(\mathbb{F}_{q^n}), \qquad \varphi(v) = tv + c_1 v^q + \cdots + c_r v^{q^r}.$$

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Definition ($\mathbb{F}_q[x]$-Drinfeld-extensiveness of rank $\leq r$)

$\mathbb{F}_{q^n}/\mathbb{F}_q$ is $\mathbb{F}_q[x]$-**Drinfeld-extensive of rank $\leq r$** if for every *cyclic* $\mathbb{F}_q[x]$-Drinfeld endomorphism φ of rank $\leq r$, there exists a primitive φ-generator $\alpha \in \mathbb{F}_{q^n}$.

- Any $\mathbb{F}_q[x]$-Drinfeld endomorphism of rank $r = 1$ is cyclic. Not the case for $r \geq 2$.
- $r \leq 1$ with $c_1 = 1 \implies \varphi = \gamma_t$ & Carlitz-extensiveness
- $r \geq n - 1$: Any $\mathbb{F}_q$-linear map is $\mathbb{F}_q[x]$-Drinfeld polynomial $\to$ Hachenberger-extensiveness

$\mathbb{F}_q[x]$-Drinfeld-Extensiveness: Classification

Hierarchy

Extensive $\Longrightarrow$ $\mathbb{F}_q[x]$-Drinfeld-extensive of rank $\leq r$ for every $r \geq 1$.

Known non-extensive pairs:

(q, n)	$\mathbb{F}_q[x]$-Drinfeld status
$(2, 2)$	Not Carlitz-ext. $\Rightarrow$ not $\mathbb{F}_q[x]$-Drinfeld-ext. rank ≤ 1
$(3, 2)$, $(5, 2)$, $(2, 4)$	Carlitz-ext., not $\mathbb{F}_q[x]$-Drinfeld-ext. rank ≤ 1
$(2, 6)$	$\mathbb{F}_q[x]$-Drinfeld-ext. rank ≤ 2, not rank ≤ 3

Extensiveness unknown:

$(2, 8)$, $(2, 10)$, $(2, 12)$, $(2, 14)$, $(2, 15)$, $(2, 16)$, $(2, 18)$, $(2, 20)$, $(2, 24)$, $(3, 8)$, $(3, 10)$, $(3, 12)$, $(4, 6)$, $(4, 9)$, $(4, 10)$, $(4, 12)$, $(5, 4)$, $(7, 6)$, $(8, 8)$.

$\mathbb{F}_q[x]$-Drinfeld-Extensiveness: Classification

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Result (Hellwig & Schaller, 2026)

We prove $(5, 4)$ is extensive, hence $\mathbb{F}_q[x]$-Drinfeld-extensive of every rank.

All remaining cases are conjectured extensive.

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Simultaneous extensiveness: matching every subfield level

New question: Find one primitive $\alpha \in \mathbb{F}_{q^n}$ that is a generator simultaneously at *every* intermediate field $\mathbb{F}_q \subseteq \mathbb{F}_{q^d} \subseteq \mathbb{F}_{q^n}$, $d \mid n$.
Note that for $d = n$, any $\alpha \neq 0$ is trivially a generator.

- Simultaneous Carlitz-extensiveness of $\mathbb{F}_{q^n}/\mathbb{F}_q$: for any $z \in \mathbb{F}_{q^n}$ for which a simultaneous $\gamma_z^{(d)}$-generator over $\mathbb{F}_{q^d}$ exists for all $d \mid n$, $d < n$, where $\gamma_z^{(d)}(v) = zv + v^{q^d}$, there exists a *primitive* simultaneous $\gamma_z^{(d)}$-generator.

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- Simultaneous $\mathbb{F}_q[x]$-Drinfeld-extensiveness of rank $\leq r$: for any $t, c_1, \ldots, c_r \in \mathbb{F}_{q^n}$ for which a simultaneous $\varphi^{(d)}$-generator over $\mathbb{F}_{q^d}$ exists for all $d \mid n$, $d < n$, where $\varphi^{(d)}(v) = tv + c_1 v^{q^d} + \cdots + c_r v^{q^{rd}}$, there exists a *primitive* simultaneous $\varphi^{(d)}$-generator.

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- Simultaneous extensiveness: for any family $(\tau_d)_{d\mid n,\, d<n}$ of $\mathbb{F}_{q^d}$-linear endomorphisms of $\mathbb{F}_{q^n}$ admitting a simultaneous generator, there exists a *primitive* simultaneous generator.
  Equivalently: Simultaneous $\mathbb{F}_q[x]$-Drinfeld-extens. with $r \geq n-1$ and coeffs not fixed.

Fixing $\varphi^{(d)}(v) = \sigma^d(v) = v^{q^d}$ recovers the Morgan–Mullen conjecture.

Three conjectures on simultaneous extensiveness

Conjecture (Simultaneous Carlitz-extensiveness)

$\mathbb{F}_{q^n}/\mathbb{F}_q$ is simultaneously Carlitz-extensive for all but finitely many pairs (q, n).

Conjecture (Simultaneous $\mathbb{F}_q[x]$-Drinfeld-extensiveness of rank $\leq r$)

For each fixed $r \geq 1$, $\mathbb{F}_{q^n}/\mathbb{F}_q$ is simultaneously $\mathbb{F}_q[x]$-Drinfeld-extensive of rank $\leq r$ for all but finitely many pairs (q, n).

Conjecture (Simultaneous extensiveness)

$\mathbb{F}_{q^n}/\mathbb{F}_q$ is simultaneously extensive for all but finitely many pairs (q, n).

Simultaneous extensiveness $\implies$ Simultaneous $\mathbb{F}_q[x]$-Drinfeld-extensiveness of rank $\leq r$
$\implies$ Simultaneous Carlitz-extensiveness

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Thank you!

Questions?

A prerequisite: from individual to simultaneous cyclicity

The simultaneous extensiveness conjectures require a simultaneous (non-primitive) generator as hypothesis. When is this guaranteed?

Conjecture (Individual $\Rightarrow$ simultaneous cyclicity NOT TRUE!)

Let $(\tau_d)_{d|n,\, d<n}$ be a family of individually cyclic $\mathbb{F}_{q^d}$-linear endomorphisms of $\mathbb{F}_{q^n}$. Then a simultaneous generator exists, i.e., there is $\alpha \in \mathbb{F}_{q^n}$ with

$$\mathbb{F}_{q^d}[\tau_d] \cdot \alpha = \mathbb{F}_{q^n} \quad \text{for all } d \mid n,\ d < n.$$

- $\tau_d = \sigma^d$: Existence of completely normal elements proven by Blessenohl–Johnsen (1986).
- $\mathbb{F}_q[x]$-Drinfeld special case $\tau_d = \varphi^{(d)}$: Fix $z, c_1, \ldots, c_r \in \mathbb{F}_{q^n}$.

$$\forall d \mid n, d < n,\ \exists \text{ a } \varphi^{(d)}\text{-generator } \alpha \in \mathbb{F}_{q^n} \overset{?}{\Longrightarrow} \exists \text{ a } \left(\varphi^{(d)}\right)_{d|n, d<n}\text{-generator } \alpha \in \mathbb{F}_{q^n}$$

- Carlitz-special case with $\varphi^{(d)} = \gamma_z^{(d)}$.